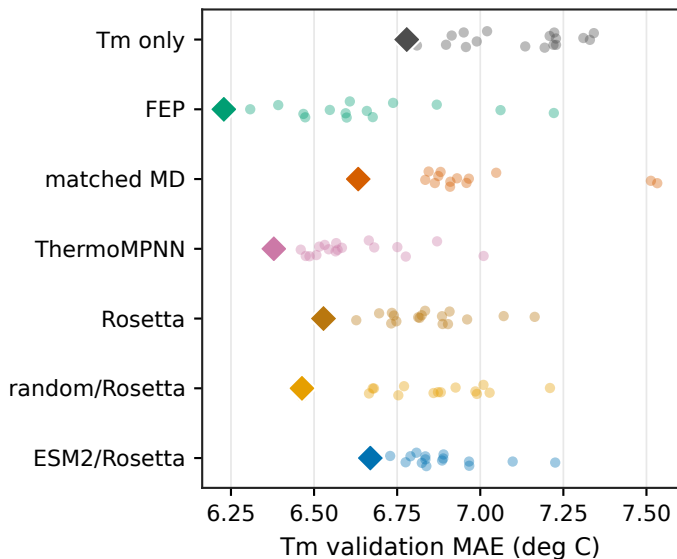
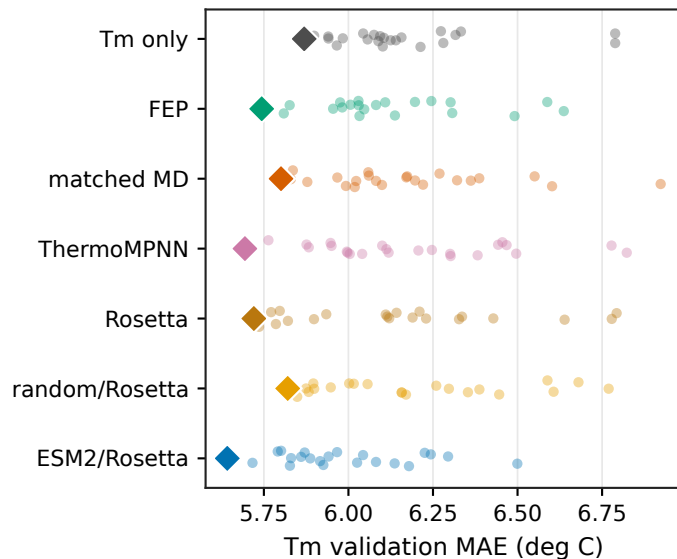


(a)

frozen ESM2

**(b)**

fine-tuned ESM2

**(c)**

Tm only	latent	-	residual	-
FEP	shared	separate	shared	separate
matched MD	shared	separate	latent	context
ThermoMPNN	shared	separate	latent	context
Rosetta	shared	separate	latent	separate
random/Rosetta	shared	separate	residual	separate
ESM2/Rosetta	shared	separate	shared	separate
	frozen arch.	frozen head	fine-tuned arch.	fine-tuned head

(d)